

# When Does Density Beat Centroids?

## A Comparative Study of Clustering Methods Across Data Geometries

[**DECIDE**: final title. Proposed: “When Clustering Fails: Which Data Break Which Methods, and Can You Tell Without Labels?”, to match the research question]

SC4020 Project 1 — Group XX

[**All**: full names exactly as on student cards; replace XX with group ID]

### Abstract

Owner: Ky (write last)

[**Ky**: 150–250 words: task, methods compared, datasets, 3 headline findings with numbers, one-line conclusion.]

[**DECIDE**: Thesis, proposed 7 Oct (Wei Yew), to match the research question in Section 1: each method fails where the data break its idea of a cluster (RQ1); the silhouette does not reveal these failures, a fake-data check reveals only some, and the rest need checks aimed at a known problem (RQ2); known remedies fix some failures at a cost, and none fixes data with no gaps (RQ3).]

## 1 Introduction

Owner: Wei Yew (draft); Ky to review

### 1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants active fault systems; a housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against.

Each clustering method carries its own idea of what a cluster is. K-means [12] looks for compact groups around a centre and assigns every point to one of them. DBSCAN [8] and HDBSCAN [5] look for dense regions separated by sparse space, and leave points in the sparse space as noise. Whether or not that idea matches the data, the method returns a partition. On one year of global earthquakes, K-means returns six tidy regions, one of which joins five plate boundaries lying 2,932 to 11,449 km apart (Section 3.4.1). Nothing in the output says that this has happened. An analyst without labels usually judges the result by an internal score such as the silhouette [17], which rates how compact and well separated the clusters are, and that score is itself built on the K-means idea of a cluster.

### 1.2 Research question

This report asks one question: *when a clustering method’s idea of a cluster does not match the data, does it fail, and can the failure be detected without labels?* We split it into three parts.

**RQ1, cause.** Which property of the data breaks which method? We consider cluster shape, density that varies between clusters, the number of dimensions, the absence of gaps between groups, feature scale, and values recorded in steps.

**RQ2, detection.** Do checks that need no labels reveal those failures? We test the silhouette, the Davies–Bouldin index, the share of points labelled noise, and a fake-data check: the

same method run on data whose features have been shuffled independently, which keeps each feature’s values but removes any joint structure.

**RQ3, remedy.** For each failure, does a known remedy help, and what does it cost?

We compare four methods, K-means with random starts, K-means++ [2], DBSCAN and HDBSCAN, on eight datasets (Table 2). Each dataset was chosen to test one data property, most of them a property that breaks at least one method’s assumption. Six have class labels, so a method’s output can be scored against them with the Adjusted Rand Index (ARI) [11], and the internal scores can be checked against that answer. Two have no labels (earthquakes, Uber pickups); there we judge failures on maps, against known geography. Gaussian mixtures, spectral clustering and agglomerative clustering appear only as reference points on the labelled datasets where we ran them (D1, D5, D6, D7), not as part of the main comparison.

### 1.3 Findings in brief

**RQ1.** Each method failed where the data broke its assumption. K-means split the two interleaved half-moons of D1 wrongly (ARI 0.48) and merged distant belts on D2. With one radius, DBSCAN clustered 90% of earthquakes in the dense south-west Pacific but only 20% in the sparser Americas. In 50 dimensions (D5) it barely matched the labels (ARI 0.03). On D8, where housing sales form one mass with no gaps, both density methods returned one dominant cluster.

**RQ2.** The silhouette did not reveal these failures. On five of the six labelled datasets, the method with the highest silhouette was not the one that matched the labels best. On D1 it ranks K-means first (0.495 against 0.392 for DBSCAN), although DBSCAN recovers the moons (ARI 0.98 against 0.48). On Ecoli (D4) it orders the four methods almost exactly opposite to ARI (Spearman  $\rho = -0.80$ ). The fake-data check, which we ran on D2 and D8, does better, but only partly. It detects when a method finds nothing real: on D8 both density methods score below the same setting on shuffled data. It does not detect the wrong structure: on D2, K-means scores well above shuffled coordinates (silhouette 0.528 against 0.391) while merging unrelated belts. Two failures were found only by checks aimed at a known problem: clusters cut at the 180° meridian, and clusters that are just one recorded storey band.

**RQ3.** Known remedies repaired some failures and not others. Measuring distance on the sphere kept all 92 close earthquake pairs across the 180° meridian together, where flat coordinates kept none. HDBSCAN raised the clustered share of the Americas from 20% to 62%, at the cost of trimming the densest belts (78% against 90%). Nothing we tried produced meaningful density clusters on D8 with all four features: capping the size of the largest cluster only made the rule select storey bands.

**[Lam, Ky, Hoang Anh: confirm the RQ2 numbers above are your settings under the shared selection rule (Section 3.3); the comment above lists each one.]**

Section 2 describes the methods and Section 3 the datasets, protocol and results, grouped by data property. Section 4 turns them into guidance.

### 1.4 Related work

**Centroid and mixture models.** Lloyd’s algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an  $O(\log k)$  approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments. Agglomerative clustering builds a hierarchy

by repeatedly merging the closest pair of clusters. Under Ward’s criterion [19] the merge cost is the increase in within-cluster variance, the same objective as K-means.

**Density-based methods.** DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius  $\varepsilon$ . Points reachable from no core point are noise. The method needs no  $k$  and imposes no shape, but its single global  $\varepsilon$  assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual reachability distances and keeps the clusters that persist longest. This removes  $\varepsilon$  and leaves a minimum cluster size as the main parameter.

**Graph-based methods.** Spectral clustering [15] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs  $O(n^2)$  memory, which limits the method to modest  $n$ .

**Validation and cluster tendency.** Internal indices, such as the silhouette [17], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [18] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

## 2 Methods

Owner: Ky (K-Means, K-Means++, DBSCAN, HDBSCAN); Lam (GMM, Spectral); Hoang Anh (Agglomerative)

### 2.1 K-Means and K-Means++

[Ky: Port Sec. 3.1.]

### 2.2 DBSCAN

[Ky: Port Sec. 3.2.]

### 2.3 HDBSCAN

[Ky: Port Sec. 3.3.]

### 2.4 Gaussian Mixture Model

[Lam: EM fit, covariance type used, parameters, complexity, assumptions.]

### 2.5 Spectral clustering

[Lam: Affinity graph, eigenvectors, K-means step; affinity used per dataset.]

## 2.6 Agglomerative clustering

[HoangAnh] Agglomerative clustering starts with every point as its own cluster and repeatedly merges the two closest clusters until  $k$  remain. The linkage defines which two are closest. We use Ward linkage [19], which merges the pair whose union raises the total within-cluster variance least. This is the K-means objective, searched greedily from the bottom up. The method therefore needs no starting centres and returns the same partition on every run, which makes it a direct check on K-means: same objective, different search. The only parameter is  $k$ , set to the number of classes (3 on D6, 2 on D7); distances are Euclidean on standardised features. It needs  $O(n^2)$  time and memory for the pairwise distances, so we ran it on the two small tables only ( $n = 178$  and  $n = 569$ ). Like K-means, it assumes compact clusters, assigns every point and has no noise label. Its own weakness is that a merge is never undone, so an early mistake stays in the final partition. On both datasets it matched the labels less well than K-means (ARI 0.790 against 0.897 on D6, 0.575 against 0.654 on D7).

**Table 1:** Structural assumptions of the methods compared.

|                            | K-Means   | K-Means++ | DBSCAN          | HDBSCAN         | GMM      | Spectral | [HoangAnh] Agglom. |
|----------------------------|-----------|-----------|-----------------|-----------------|----------|----------|--------------------|
| Cluster shape assumed      | isotropic | isotropic | arbitrary       | arbitrary       | [Lam: ?] | [Lam: ?] | compact (Ward)     |
| Requires $k$               | yes       | yes       | no              | no              | [Lam: ?] | [Lam: ?] | yes                |
| Explicit noise label       | no        | no        | yes             | yes             | [Lam: ?] | [Lam: ?] | no                 |
| Handles varying density    | no        | no        | no              | yes             | [Lam: ?] | [Lam: ?] | no                 |
| Parameter is interpretable | no        | no        | yes (km)        | partly          | [Lam: ?] | [Lam: ?] | no                 |
| Time complexity            | $O(nkdi)$ | $O(nkdi)$ | $O(n \log n)^a$ | $O(n \log n)^a$ | [Lam: ?] | [Lam: ?] | $O(n^2)$           |

<sup>a</sup> With an effective spatial index;  $O(n^2)$  worst case.

## 3 Experiments

### 3.1 Datasets

Owner: each dataset’s owner writes its paragraph (max. 120 words)

**Table 2:** Datasets, grouped by the data property each one tests. [DECIDE: final list; drop or keep Mall; Ky’s 1965–2016 earthquakes replaced by the 2023 surface-distance pipeline?]

| ID | Dataset                         | Owner          | $n$                 | $d$ | Labels        | Property tested               |
|----|---------------------------------|----------------|---------------------|-----|---------------|-------------------------------|
| D1 | make_moons (synthetic)          | Lam            | 1,000               | 2   | 2 classes     | curved shapes (control)       |
| D2 | Earthquakes, 2023, $M \geq 4.5$ | Wei Yew        | 7,643               | 2   | none          | long belts, uneven density    |
| D3 | Uber NYC pickups, Apr 2014      | Ky, Hoang Anh  | [DECIDE: 6k or 15k] | 2   | none          | density contrast in a city    |
| D4 | Ecoli                           | Ky             | 336                 | 7   | 8 classes     | dimensionality, rare classes  |
| D5 | Fashion-MNIST (PCA)             | Lam            | 3,000               | 50  | 10 classes    | high dimensions               |
| D6 | Wine                            | Lam, Hoang Anh | 178                 | 13  | 3 classes     | silhouette vs labels          |
| D7 | Breast cancer (Wisconsin)       | Hoang Anh      | 569                 | 30  | 2 classes     | silhouette vs labels          |
| D8 | HDB resale flats, 2017–2026     | Wei Yew        | 20,000 <sup>b</sup> | 4   | flat type (7) | no gaps, scale, recorded ties |

<sup>b</sup> Random sample of 234,294 usable sales; all 234,294 used for runtime (Section 3.7).

**D1 make\_moons.** [Lam: generator settings (noise 0.07), why it is the shape control.]

**D2 Earthquakes.** Every event of magnitude 4.5 or more in 2023 from the USGS ComCat catalogue [21]: 7,651 events, of which 8 volcanic eruptions near Saipan are removed, leaving 7,643 earthquakes. We cluster the epicentre only. Depth is left out because 44.2% of events carry the default depth of 10 km. Each method meets a different difficulty. Earthquakes lie along narrow belts thousands of kilometres long, which breaks the compact-cluster assumption of K-means. Density varies widely: the distance to the nearest earthquake runs from 1.2 km (5th percentile) to 100.5 km (95th), so no single DBSCAN radius suits every belt. And 17.2% of earthquakes lie within  $10^\circ$  of the  $180^\circ$  meridian, where flat coordinates cut the map. There are no labels.

**D3 Uber NYC.** [Ky + Hoang Anh: one merged paragraph; agree sample size and bounding box.]

**D4 Ecoli.** [Ky: Port from Sec. 4.1.]

**D5 Fashion-MNIST.** [Lam: 3,000-image sample, PCA to 50 dims; why subsampled.]

**D6 Wine.** [HoangAnh] The UCI Wine data [23]: 178 wines grown in one Italian region, each described by 13 chemical measurements such as alcohol, malic acid, flavanoids and proline. The cultivar (3 classes of 59, 71 and 48 wines) is held out as the label. Features are standardised, because proline runs from 278 to 1,680 while hue stays below 2. The classes are compact, so K-means recovers them well: two members ran it independently and both obtained ARI 0.897. The dataset is in the study for the opposite reason. No sparse gap separates the cultivars, so the density methods return 2 clusters with about a third or more of the wines as noise, and the silhouette still prefers them (Section 3.4.5).

**D7 Breast cancer.** [HoangAnh] The Wisconsin diagnostic breast cancer data [24]: 569 tumours, each described by 30 measurements of the cell nuclei in an image of a fine-needle aspirate (the mean, standard error and largest value of 10 properties such as radius, texture and concavity). The diagnosis, 357 benign and 212 malignant, is held out as the label. Features are standardised, because areas reach 4,254 while smoothness and symmetry stay below 1. Two properties make the data hard. The classes differ in spread: malignant tumours lie 35% further from their class centre than benign ones (5.47 against 4.06 in standardised units), which breaks the equal-size assumption of K-means. And the two classes form one connected mass in 30 dimensions, with no sparse gap for a density method to cut along.

**D8 HDB resale.** Every resale of a public housing flat in Singapore from January 2017, from data.gov.sg [22]. The 6,051 sales after June 2026 (2.5%) have no price index value yet and are dropped, leaving 234,294. Methods are compared on a random sample of 20,000 (seed 0); all 234,294 are used for runtime. There are four features: floor area, remaining lease, storey, and price adjusted to 2026 levels with the HDB resale price index. Flat type (7 classes) is held out as a label, but it is nearly a function of size: floor area alone reaches ARI 0.601 against it. The data are hard for density methods. There are no gaps between groups of flats, and storeys are recorded in 3-storey bands, so 77.7% of sales share their floor area, lease and storey with at least one other sale.

## 3.2 Preprocessing

Owner: Wei Yew

Three rules apply wherever a dataset has the property they address. Each is tested by an ablation in Section 3.5.

**Coordinates on the sphere.** Latitude and longitude treated as flat numbers put two earthquakes 6.07 km apart at 359.94° of longitude from each other, because they lie on either side of the 180° meridian. Density methods therefore use the haversine (great-circle) distance on coordinates in radians. K-means needs a vector space to average in, so each epicentre becomes a 3D point on the unit sphere. The straight-line distance between two such points grows with their great-circle distance, so the clustering follows surface distance (Ablation B).

**Standardised features.** Features in their raw units are not comparable. On D8, price (in dollars) accounts for 100.0% of the squared distance between sales, so the other three features are ignored. Every feature is scaled to mean 0 and standard deviation 1 (Ablation C).

**Values recorded in steps.** When a feature is recorded in steps, many points share exact values, and density methods read each shared value as a dense spot. Before running a density method on D8, each stepped value is moved by a uniform random amount of up to half its step:  $\pm 1.5$  storeys,  $\pm 0.5 \text{ m}^2$  of floor area and  $\pm \frac{1}{24}$  year of lease. Price is continuous and is left unchanged. Results are reported with and without this jitter (Ablation E).

### 3.3 Evaluation metrics and parameter selection

Owner: Ky, Wei Yew

**Metrics.** Every result table reports the same columns. *Internal* scores need no labels. The silhouette coefficient [17] of a point is  $(b - a) / \max(a, b)$ , where  $a$  is its mean distance to its own cluster and  $b$  its mean distance to the nearest other cluster; we report the mean over points, from  $-1$  to  $1$ , higher is better. The Davies–Bouldin index [6] averages, over clusters, the worst ratio of within-cluster spread to between-centre distance; lower is better. Both are computed on clustered points only, so they say nothing about the points a density method calls noise. Two *coverage* figures therefore stand next to them: the noise fraction and the share of points in the largest cluster. *External* scores need labels. The Adjusted Rand Index (ARI) [11] counts pairs of points on which a clustering and the labels agree, corrected so that a random clustering scores about  $0$  and a perfect one  $1$ . Normalised mutual information (NMI) measures the same agreement in terms of shared information. For both, noise counts as one more cluster, so a method cannot raise its score by discarding hard points. We also report the fit time.

**Fake-data check.** An internal score has no fixed scale: a silhouette of  $0.3$  can be strong on one dataset and meaningless on another. Each chosen setting is therefore run again on fake data that keeps each feature’s distribution but has no joint structure. For tables, each column is shuffled independently. For coordinates, points are drawn uniformly in the same latitude and longitude range. A result counts as structure only if the real data score clearly above the fake data.

**Selection rule.** One rule picks the setting of DBSCAN and HDBSCAN on every dataset: among the settings that give at least 2 clusters and label at most 50% of points as noise, take the one with the highest silhouette. The noise cap is needed because the silhouette ignores noise. Without it, the silhouette prefers settings that throw most of the data away. On D2 it would pick DBSCAN with  $\varepsilon = 20.4 \text{ km}$ , which scores  $0.878$  by calling 69.4% of earthquakes noise. On D8 it would pick HDBSCAN with `min_cluster_size` 400, which scores  $0.448$  with 70.9% of sales as noise. For K-means,  $K$  is chosen from the elbow in inertia, the silhouette, the gap to the fake-data silhouette and agreement between seeds; each dataset states which of these decided. On labelled data we report the rule’s choice and the setting with the best ARI, so the cost of choosing without labels is visible.

**Parameter grids.** K-means and K-means++:  $K = 2$  to  $12$ , 10 starts each. DBSCAN: `minPts` (`min_samples` in scikit-learn) in  $\{5, 10, 20\}$ . Its radius  $\varepsilon$  comes from the k-distance curve, the sorted distance from each point to its `minPts`-th neighbour: on D2 we take the 25th, 50th, 75th, 90th and 95th percentiles of the curve (11 to 810 km); on D8 we take 0.5, 0.75, 1, 1.5 and 2 times the knee of the curve. HDBSCAN: `min_cluster_size` scaled to the data (25, 50, 100 on D2; 100, 200, 400 on D8) and `min_samples` in  $\{\text{default}, 5, 10\}$ . [HoangAnh] On D6 and D7, K-means is searched over  $K = 2$  to  $10$  with 10 starts and reported at the number of classes. GMM takes 2 to 10 components and four covariance types (full, tied, diagonal, spherical), chosen by BIC. DBSCAN takes `minPts` in  $\{10, 20, 40, 80\}$  and  $\varepsilon$  at 0.5 to 4 times the knee of the k-distance curve. HDBSCAN takes `min_cluster_size` from 5 up to  $n/10$  (15 on D6, 50 on D7) and `min_samples` in  $\{\text{default}, 5, 10\}$ . On these two datasets the density settings that pass the constraints (at least 2 clusters,

at most 50% noise, and at most 10 clusters) are ranked by DBCV [14], a score built for density clusters, instead of the silhouette. Agglomerative clustering uses Ward linkage at the number of classes. **[Ky, Lam: one line each for your datasets, and GMM components and spectral affinity.]**

**Reproducibility.** On D2 and D8 every random step uses seed 0, and repeated runs use seeds 0 to 19. They ran on Python 3.12.13, scikit-learn [16] 1.9.1 and NumPy 2.5.3 on an Apple M2 with 16 GB of memory. **[Ky, Lam: versions and hardware if different.]** **[Hoang Anh: D6, D7 ran on Python 3.13 and scikit-learn 1.9; fill in exact versions and hardware.]** **[DECIDE: shared code: Ky’s src/core.py or Wei Yew’s metrics.py, so every table has the same columns.]**

### 3.4 Comparison by data property

Owner: All

**[All: One results table per property (rows: dataset  $\times$  method; columns: #clusters, noise, S, DB, ARI, NMI, time, fake-data gap). Bold the best per dataset.]**

**Table 3:** D2 and D8 at the settings chosen in Section 3.3. S: silhouette; DB: Davies–Bouldin; null S: silhouette of the same setting on fake data (D2: shuffled coordinates for K-means, uniform points for density methods; D8: shuffled columns). “none”: no clusters formed on the fake data. D2 has no labels.

| Data | Method                | Clusters | Noise | Largest | S     | DB    | ARI   | NMI   | Time (s) | Null S |
|------|-----------------------|----------|-------|---------|-------|-------|-------|-------|----------|--------|
| D2   | K-Means ( $K = 6$ )   | 6        | 0%    | 44.3%   | 0.525 | 0.800 | –     | –     | 0.004    | 0.389  |
| D2   | K-Means++ ( $K = 6$ ) | 6        | 0%    | 31.0%   | 0.528 | 0.738 | –     | –     | 0.005    | 0.391  |
| D2   | DBSCAN (101.8 km, 20) | 41       | 40.4% | 11.8%   | 0.703 | 0.351 | –     | –     | 0.051    | none   |
| D2   | HDBSCAN (50)          | 28       | 30.7% | 7.5%    | 0.757 | 0.329 | –     | –     | 1.081    | 0.320  |
| D8   | K-Means ( $K = 4$ )   | 4        | 0%    | 36.2%   | 0.317 | 1.116 | 0.220 | 0.299 | 0.011    | 0.195  |
| D8   | K-Means++ ( $K = 4$ ) | 4        | 0%    | 35.1%   | 0.317 | 1.116 | 0.213 | 0.293 | 0.010    | 0.218  |
| D8   | DBSCAN (0.567, 10)    | 3        | 1.4%  | 97.1%   | 0.239 | 0.732 | 0.030 | 0.091 | 0.428    | 0.330  |
| D8   | HDBSCAN (200)         | 2        | 28.5% | 68.6%   | 0.160 | 1.133 | 0.072 | 0.101 | 2.022    | 0.593  |

#### 3.4.1 Curved and elongated clusters

Owner: Lam (D1), Wei Yew (D2)

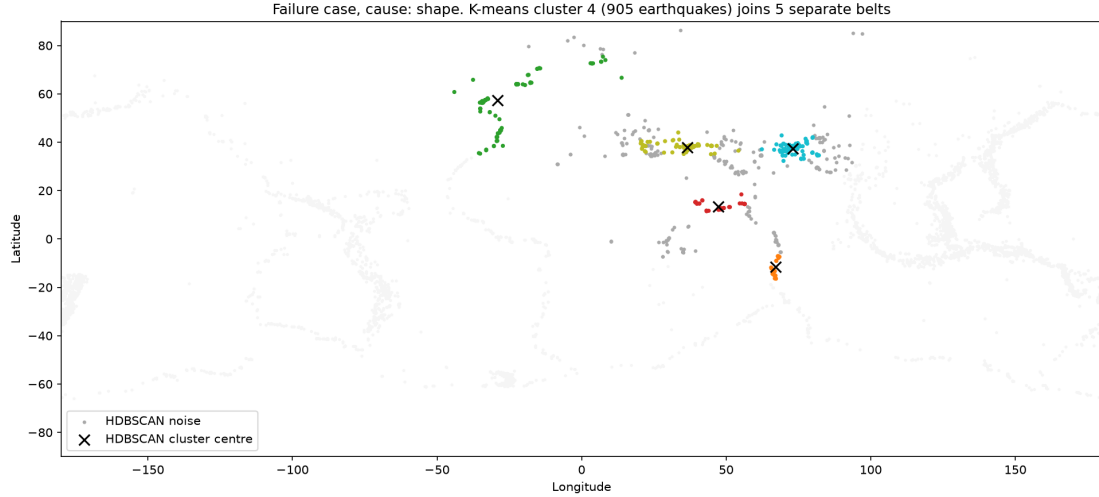
**[Lam: D1: DBSCAN ARI 0.982, Spectral 1.000, K-Means++ 0.481 (lam/results/make\_moons\_results.csv).]**

K-means assumes compact clusters, and D2 breaks that assumption at planetary scale. Earthquakes follow plate boundaries, so a cluster is a long, narrow belt rather than a ball around a centre. K-Means++ with  $K = 6$  must cover the whole globe with six roughly round regions, so each region takes in several unrelated belts (Figure 1). One cluster of 905 earthquakes joins Turkey, the Reykjanes Ridge and Iceland, Afghanistan and Tajikistan, Somalia and Eritrea, and the Mid-Indian Ridge. HDBSCAN finds each of these as its own cluster, and their centres are 2,932 to 11,449 km apart. K-means also has no noise label, so every isolated earthquake in the oceans is assigned to some region. Its silhouette is still 0.528, well above the 0.391 of shuffled coordinates: the score confirms that the data are not random, not that the clusters are meaningful.

#### 3.4.2 Uneven density

Owner: Wei Yew (D2), Ky and Hoang Anh (D3)

On D2 the distance to the nearest earthquake runs from 1.2 km to 100.5 km (5th to 95th percentile), and the k-distance curve rises smoothly with no knee. Any single  $\varepsilon$  is therefore too small for some belts and too large for others. At the chosen  $\varepsilon = 101.8$  km, DBSCAN clusters 90% of the earthquakes in the dense south-west Pacific but only 20% in the sparser Americas (Figure 2).



**Figure 1:** D2, failure of K-Means++ ( $K = 6$ ). All points shown in colour or dark grey belong to one K-means cluster; colours are the HDBSCAN clusters inside it, crosses their centres. Cause: cluster shape. Belts thousands of kilometres apart are joined because K-means can only draw compact regions.

HDBSCAN, which picks a density level per cluster, clusters 62% of the Americas and keeps the Mexico to Panama belt and the Chile and Argentina belt as long clusters. The price is paid in the dense region, where it clusters 78% instead of 90%. Overall HDBSCAN keeps more earthquakes (69.3% against 59.6%) with a higher silhouette (0.757 against 0.703, Table 3).

[Ky + Hoang Anh: D3: Midtown vs outer boroughs; knee  $\varepsilon$  vs silhouette-optimal  $\varepsilon$ .]

### 3.4.3 High dimensions

Owner: Ky (D4), Lam (D5)

[Ky: D4: crossover between 3 and 4 PCA dimensions (Ablation D).] [Lam: D5: DBSCAN ARI 0.033; HDBSCAN labels 71% of points noise.]

### 3.4.4 No gaps between groups

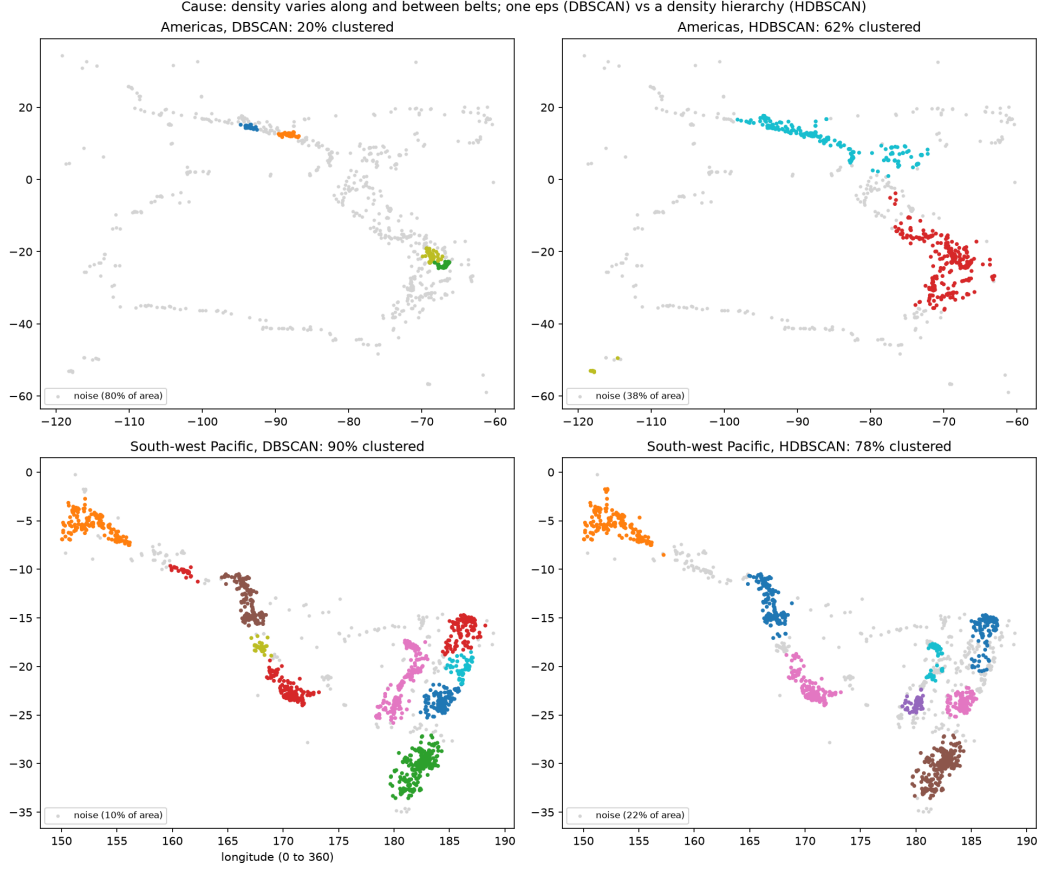
Owner: Wei Yew (D8)

D8 is one continuous mass: flats come in every combination of size, lease, storey and price, with no empty space between groups. Table 3 shows what each method does with it. K-means still finds a weak partition, four market segments (newer mid-size, older small, large older, newer high-floor), with a silhouette of 0.317 against 0.218 on shuffled columns. That ratio of 1.45 is modest, but it is the only result on D8 that beats its fake data. Density methods have no sparse region to cut along. The DBSCAN setting chosen by the rule puts 97.1% of sales in one cluster, and HDBSCAN finds one core holding 68.6% of sales plus a compact block of new 3-room flats, and calls 28.5% noise. Both score below the same setting on shuffled columns (0.239 against 0.330, and 0.160 against 0.593). On shuffled columns HDBSCAN calls 94.0% of sales noise and keeps two small, well-separated clusters, which the silhouette rewards. On data with no gaps, a density method has nothing to find, and the selection rule cannot tell.

### 3.4.5 Silhouette versus ground truth

Owner: All

[All: One paragraph, one line per dataset: D4 best silhouette (HDBSCAN) has the worst ARI, rank correlation  $-0.80$  (Ky); D6 DBSCAN silhouette 0.42 vs K-Means++ 0.28, ARI 0.31 vs 0.90

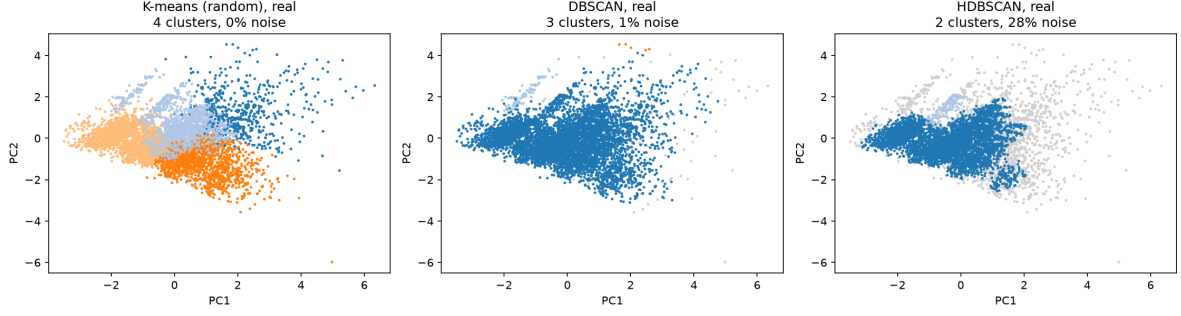


**Figure 2:** D2, DBSCAN ( $\epsilon = 101.8$  km) against HDBSCAN in a sparse region (top, the Americas) and a dense one (bottom, the south-west Pacific); grey points are noise. Success of HDBSCAN on the sparse belts, at the cost of trimming the dense ones. Cause: density varies along and between belts.

(Lam); D8 unscaled K-means silhouette 0.540 from price bands, against 0.317 standardised (Wei Yew, written in Ablation C); D2 K-means silhouette flat at 0.516 to 0.532 for every  $K$  from 4 to 12, so it cannot choose  $K$  (Wei Yew, [weiyew/sc4020-earthquake/results/kmeans\\_k\\_selection.csv](https://github.com/WeiYew/sc4020-earthquake/blob/main/results/kmeans_k_selection.csv)); D5 HDBSCAN top silhouette with 71% noise (Lam).]

**D7 Breast cancer.** [HoangAnh] Given  $k = 2$ , the four methods that assign every point score almost the same silhouette, 0.314 to 0.343, yet their ARI runs from 0.575 to 0.774 (Table 4). The two scores disagree at the top: K-means has the highest silhouette of the four and GMM the lowest, while GMM matches the diagnosis best. The silhouette rewards compact clusters of similar size. The malignant class is more spread out than the benign one (Section 3.1), and a GMM with a full covariance per component can fit that difference where K-means cannot. Agglomerative clustering with Ward linkage optimises the K-means objective greedily and does worst of the four (ARI 0.575). The density methods fail outright. None of the 24 settings tried gives at least 2 clusters with at most 50% noise. The least noisy two-cluster settings discard 55.5% (DBSCAN) and 59.2% (HDBSCAN) of the tumours, and on what remains they score the two highest silhouettes of any method, 0.356 and 0.364, with the two lowest ARI, 0.222 and 0.156. Without the noise cap of Section 3.3, the silhouette would rank the two worst results first.

**D6 Wine and the fake-data check.** [HoangAnh] D6 repeats the pattern inside the noise cap. DBSCAN and HDBSCAN pass the cap with about a third of the wines as noise, take the two highest silhouettes (0.356 and 0.352) and the two lowest ARI (0.296 and 0.342), while GMM and K-means reach 0.915 and 0.897 with silhouettes near 0.28. The fake-data check does not separate



**Figure 3:** D8 at the chosen settings on recorded data, drawn on the first two principal components (6,000 sales; grey: noise). K-means cuts the mass into four segments; DBSCAN and HDBSCAN return one dominant cluster. Cause: no gaps between groups.

good from bad here either. Every method that assigns all points scores far above its shuffled-column silhouette on both datasets (0.285 against 0.066 for K-means on D6, 0.343 against 0.033 on D7), and the density methods form no cluster at all on shuffled data. So each result passes the check, including the density results that match the labels worst: what they find, a dense core of one class, is real structure, but it is not the class structure. The check shows that a result is not an artefact of the marginals. It does not show which of several real structures a method has found.

**Table 4:** [HoangAnh] D6 and D7 in the shared columns. S: silhouette; DB: Davies–Bouldin, both on clustered points only; null S: silhouette of the same setting on shuffled columns (“none”: no clusters formed). Every fit takes under 0.3 s. On D7 the two density rows break the 50% noise cap and are shown for comparison only. Bold: best S and best ARI per dataset.

| Data | Method                         | Clusters | Noise | Largest | S            | DB    | ARI          | NMI   | Null S |
|------|--------------------------------|----------|-------|---------|--------------|-------|--------------|-------|--------|
| D6   | K-Means ( $k = 3$ )            | 3        | 0%    | 36.5%   | 0.285        | 1.389 | 0.897        | 0.876 | 0.066  |
| D6   | Agglomerative (Ward, $k = 3$ ) | 3        | 0%    | 36.0%   | 0.277        | 1.419 | 0.790        | 0.786 | 0.045  |
| D6   | Spectral ( $k = 3$ )           | 3        | 0%    | 36.0%   | 0.283        | 1.391 | 0.880        | 0.861 | 0.038  |
| D6   | GMM (3, diagonal)              | 3        | 0%    | 39.3%   | 0.280        | 1.412 | <b>0.915</b> | 0.892 | 0.054  |
| D6   | DBSCAN (2.33, 10)              | 2        | 33.7% | 49.4%   | <b>0.356</b> | 1.062 | 0.296        | 0.391 | none   |
| D6   | HDBSCAN (5)                    | 2        | 32.6% | 48.3%   | 0.352        | 1.093 | 0.342        | 0.430 | none   |
| D7   | K-Means ( $k = 2$ )            | 2        | 0%    | 65.9%   | 0.343        | 1.321 | 0.654        | 0.532 | 0.033  |
| D7   | Agglomerative (Ward, $k = 2$ ) | 2        | 0%    | 67.7%   | 0.339        | 1.370 | 0.575        | 0.457 | 0.018  |
| D7   | Spectral ( $k = 2$ )           | 2        | 0%    | 67.0%   | 0.337        | 1.316 | 0.761        | 0.663 | 0.023  |
| D7   | GMM (2, full)                  | 2        | 0%    | 62.4%   | 0.314        | 1.377 | <b>0.774</b> | 0.661 | 0.043  |
| D7   | DBSCAN (2.33, 10)              | 2        | 55.5% | 42.0%   | 0.356        | 0.853 | 0.222        | 0.255 | none   |
| D7   | HDBSCAN (5)                    | 2        | 59.2% | 38.5%   | <b>0.364</b> | 0.844 | 0.156        | 0.213 | none   |

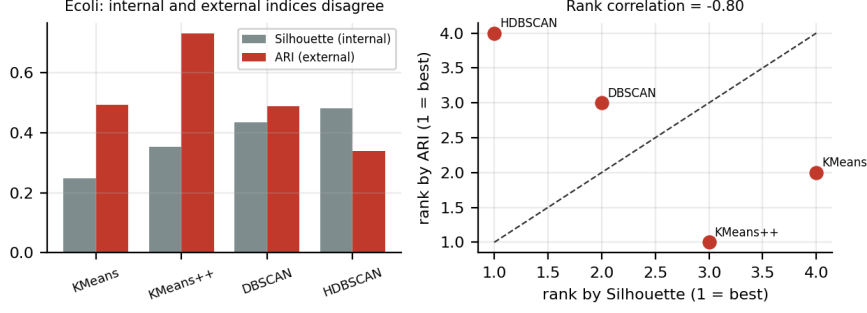
### 3.5 Ablation studies

Owner: Ky, Wei Yew

#### 3.5.1 A: Initialisation (K-Means vs K-Means++)

[Ky: Port Ablation A (30 seeds, D3/D4). Add ARI mean  $\pm$  s.d. over the 30 seeds: the 0.24 ARI claim currently rests on seed 42 only.]

On D2 and D8 each initialisation ran from one start with 20 seeds at the chosen  $K$ ; we measure how far each run’s inertia lies above the best run. On D2 ( $K = 6$ ) K-Means++ helps clearly: the mean gap falls from 10.9% to 4.8%, the worst from 18.8% to 11.5%, and iterations from 15.3 to 10.3. It does not remove the need for restarts. Two K-Means++ runs still agree only at ARI 0.80 on average, against 0.78 for random starts, because the belts give many local optima of similar inertia. On D8 ( $K = 4$ ) the choice does not matter: all 40 runs end within 0.009% of the best



**Figure 4:** [Ky: existing figure; extend to D5–D8 or make one panel per dataset.]

inertia. Initialisation matters when the objective has many local optima of different quality; on D8 every start reaches practically the same one.

### 3.5.2 B: Distance on geographic coordinates

Every method ran at its chosen setting twice on D2: once on the sphere (Section 3.2) and once on latitude and longitude as flat numbers (DBSCAN’s  $\varepsilon$  converted at 111.2km per degree). To isolate the effect of the 180° meridian we count *cross-line pairs*: the 92 pairs of earthquakes within 50km of each other on opposite sides of the line. Table 5 shows that flat coordinates separate every one of these pairs, for every method (Figure 5). The Fiji–Tonga–Kermadec system is cut in two. K-Means++ changes everywhere: its silhouette falls from 0.528 to 0.404 and its partition agrees with the sphere partition at ARI 0.557 only. DBSCAN hides the failure in its global numbers. Its silhouette moves from 0.703 to 0.701 and its partition still agrees at ARI 0.948, because only the clusters that cross the line are split. A check aimed at the known problem, here the cross-line count, finds what the internal scores miss.

**Table 5:** D2 on the sphere and on flat degrees. Cross-line pairs: 92 close pairs that straddle the 180° meridian; “kept” counts pairs placed in the same cluster.

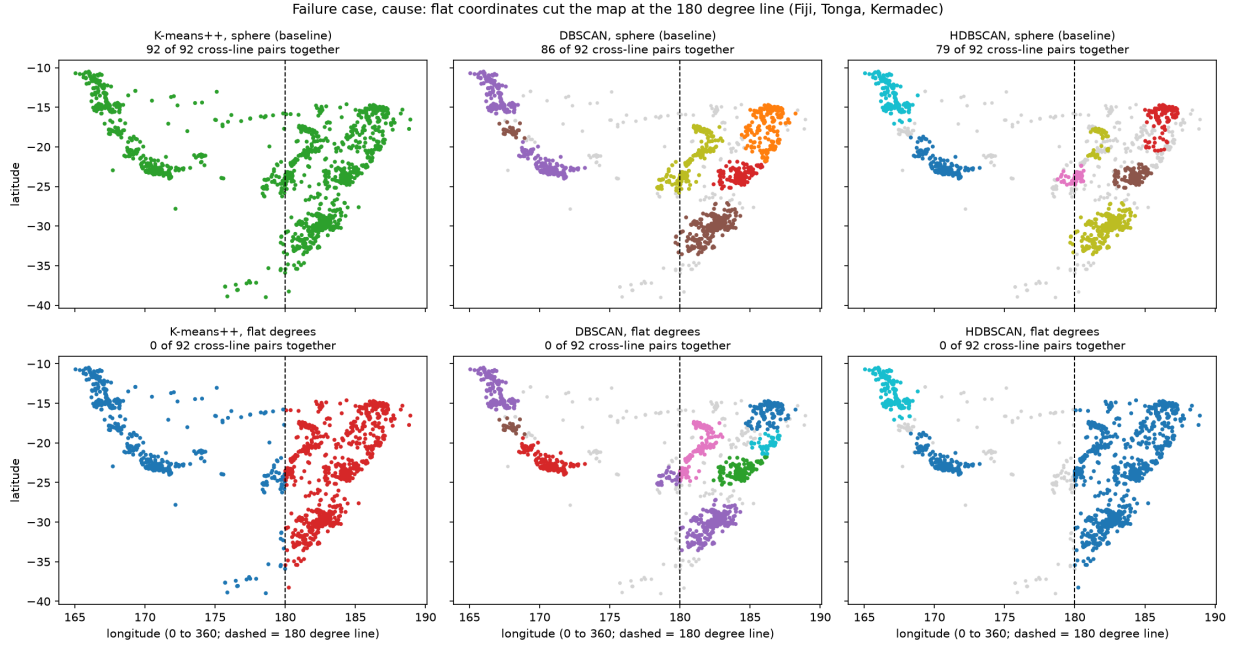
| Method    | Distance | Clusters | Silhouette | ARI vs sphere | Pairs kept |
|-----------|----------|----------|------------|---------------|------------|
| K-Means++ | sphere   | 6        | 0.528      | 1.000         | 92 of 92   |
|           | flat     | 6        | 0.404      | 0.557         | 0 of 92    |
| DBSCAN    | sphere   | 41       | 0.703      | 1.000         | 86 of 92   |
|           | flat     | 43       | 0.701      | 0.948         | 0 of 92    |
| HDBSCAN   | sphere   | 28       | 0.757      | 1.000         | 79 of 92   |
|           | flat     | 23       | 0.721      | 0.784         | 0 of 92    |

[DECIDE: Ky’s projected-km pipeline has the same  $\pm 180^\circ$  cut. Rerun D3 on the sphere, or state that city scale makes it irrelevant.]

### 3.5.3 C: Feature scaling

[Ky: D4: standardisation lifts K-Means++ ARI 0.497  $\rightarrow$  0.731; DBSCAN unaffected once  $\varepsilon$  is re-tuned.]

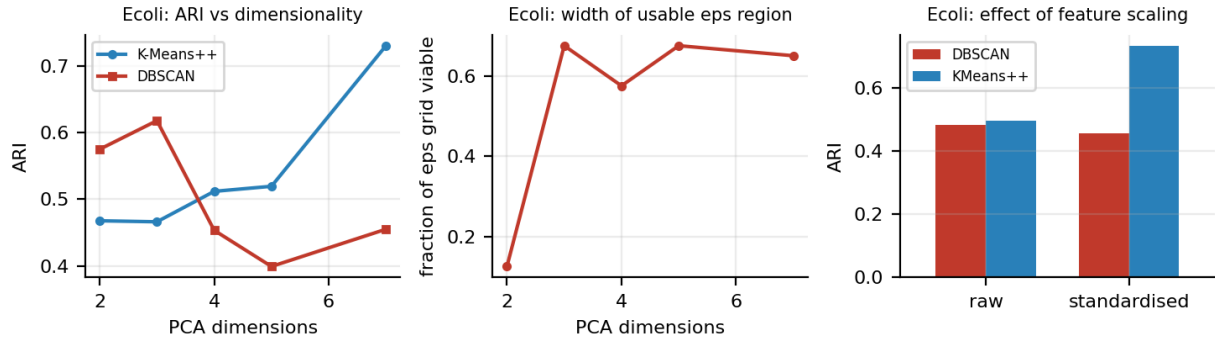
On D8, K-means ( $K = 4$ ) on raw units scores a silhouette of 0.540, well above the 0.317 of the standardised run. The higher score is not a better clustering. In dollars, price makes up 100.0% of the squared distance between sales, so the four clusters are price bands and floor area, lease and storey play no part. The raw partition agrees with the standardised one at ARI 0.261 only, and neither matches flat type (ARI 0.229 raw, 0.214 standardised). Scaling decides which question the clustering answers. The silhouette, measured in each run’s own units, cannot say which is right.



**Figure 5:** Fiji, Tonga and Kermadec, drawn on longitudes 0 to 360° (dashed: the 180° meridian). Top: on the sphere, clusters cross the line. Bottom: on flat degrees, every method splits the system at the line. Cause: flat coordinates treat 179.9°E and 179.9°W as 359.8° apart.

### 3.5.4 D: Dimensionality

[Ky: Port Ablation D. Call it PCA dimension; “intrinsic dimensionality” overclaims.]

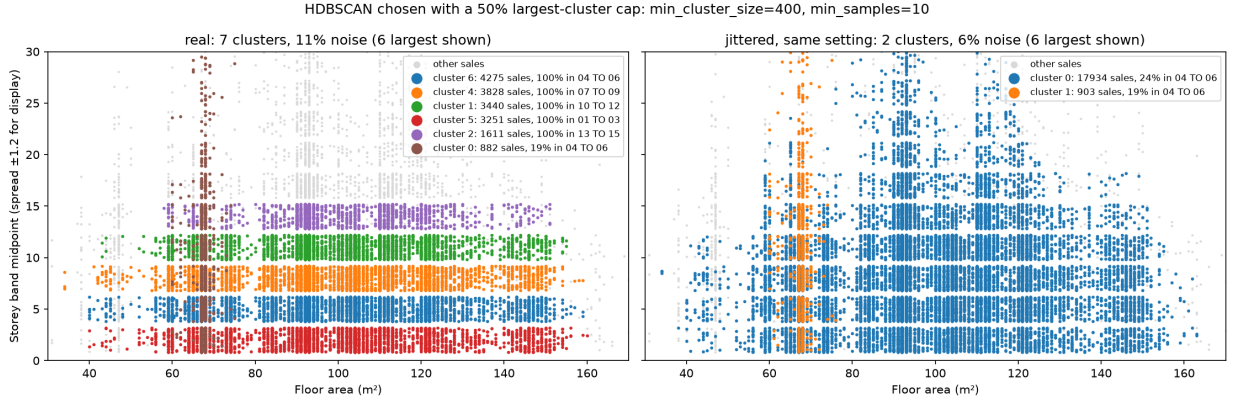


**Figure 6:** [Ky: existing figure (Ablations C and D on D4).]

### 3.5.5 E: Values recorded in steps

On D8 the selection rule picks density settings dominated by one cluster: it holds 97.1% of sales for DBSCAN, and 68.6% for HDBSCAN, which calls another 28.5% noise (Section 3.4.4). A natural repair is to add a cap on the largest cluster. We tested caps from 90% down to 50% on the data as recorded and on jittered data (Section 3.2). On recorded data a cap of 50% does give balanced partitions, but of the wrong kind: all 27 DBSCAN clusters, and 6 of the 7 HDBSCAN clusters, each contain a single storey band (Figure 7). They are stacks of identical recorded values, not groups of similar flats. On jittered data, where those ties are spread out, *no* setting meets a cap of 50% or 60%. The balanced partitions existed only because of the recording steps. At the uncapped HDBSCAN setting, by contrast, recorded and jittered data agree (ARI 0.886), so the chosen setting is not a tie artefact. Two lessons follow. Stepped values must be checked before a density method

runs. And a constraint added to a selection rule can make it pick artefacts, which is why we kept the rule unchanged.



**Figure 7:** D8, HDBSCAN chosen under a 50% largest-cluster cap. Left, recorded values: 6 of 7 clusters are single storey bands. Right, the same setting on jittered values: 2 clusters. Cause: storeys recorded in 3-storey bands create exact ties that density methods read as dense regions.

### 3.5.6 F: Data subsets

Each D2 subset is compared with a random subset of the same size, so that an effect of the subset is not confused with an effect of size. Clustering January to June or July to December separately, K-Means++ reproduces its full-year labels at ARI 0.963 and 0.829 (random halves: 0.968 and 0.994); the drop for the second half comes from aftershock sequences. Density methods change as much on random halves as on the real ones (HDBSCAN 0.414 and 0.472, against 0.505 and 0.544). Their parameters are counts of points, so their output depends on how many points there are. Keeping only magnitude 5.0 and above (1,780 earthquakes) behaves like a random subset of the same size (DBSCAN ARI 0.343 against 0.315, HDBSCAN 0.322 against 0.303). Larger earthquakes do not outline the belts more clearly, so we keep the 4.5 threshold.

## 3.6 Improvements to the methods

Owner: All

**Table 6:** Implemented improvements. Each fix targets a failure observed in Section 3.4.

| Failure                              | Cause                    | Fix                           | Data   | Before → after                        | Owner       |
|--------------------------------------|--------------------------|-------------------------------|--------|---------------------------------------|-------------|
| Clusters split at 180°               | flat lat/lon             | surface distance              | D2     | 0 → 92 of 92 pairs kept               | Wei Yew     |
| Poor local optimum                   | random seeding           | K-Means++                     | D4     | ARI 0.493 → 0.731                     | Ky          |
| Sparse belts lost                    | one global $\varepsilon$ | HDBSCAN                       | D2     | Americas kept 20% → 62%               | Wei Yew     |
| One feature dominates                | unequal scales           | standardisation               | D4     | ARI 0.497 → 0.731                     | Ky          |
| Clusters are storey bands            | recorded steps           | jitter within step            | D8     | band clusters 107 of 124 → 0 of 23    | Wei Yew     |
| Curved clusters cut                  | convex assumption        | spectral clustering           | D1     | ARI 0.481 → 1.000                     | Lam         |
| [HoangAnh] Classes of unequal spread | equal-size assumption    | GMM, full covariance          | D7     | ARI 0.654 → 0.774                     | Hoang Anh   |
| Degenerate $k$ chosen                | silhouette alone         | constraints + fake-data check | D2, D3 | D2 null gap 0.015 → 0.144; D3 [Ky: ?] | Ky, Wei Yew |

**Surface distance (D2).** Haversine distance for density methods and 3D unit vectors for K-means measure what matters, the distance along the Earth’s surface. The fix is exact, and its only cost is that  $\varepsilon$  must be given in kilometres and the coordinates converted. It changes the answer for every method (Ablation B), which makes it a correctness fix rather than a tuning choice.

**HDBSCAN instead of DBSCAN (D2).** HDBSCAN replaces the single radius with a hierarchy of densities and keeps the clusters that persist longest, so sparse and dense belts can both survive. In the Americas it clusters 62% of earthquakes against 20%. The costs are real: dense cores are

trimmed (78% against 90% in the south-west Pacific), each fit takes about 20 times longer (1.08 s against 0.05 s), and `min_cluster_size` is a count, so the result changes with the size of the dataset (Ablation F).

**Jitter within the recording step (D8).** Moving each stepped value by up to half its step spreads exact ties without moving any sale into another band. Across the 9 HDBSCAN settings, 107 of the 124 clusters found on recorded data are single storey bands; on jittered data, 0 of 23 are. The fix costs nothing at the chosen setting (recorded and jittered agree at ARI 0.886), but it adds randomness, so the seed must be fixed and reported. It also shows that D8 has fewer real density clusters than the recorded data suggest.

**Fake-data check for choosing  $K$  (D2).** The silhouette alone picks  $K = 2$  on D2 (0.544, the highest of any  $K$ ), but shuffled coordinates score 0.529 at the same  $K$ : the split follows the distribution of latitude and longitude on their own, not their combination. Requiring a clear gap over the fake data rules out  $K = 2$  and  $K = 3$ . From  $K = 4$  the gap is 0.126 to 0.149 and almost flat, so the final choice,  $K = 6$ , rests on agreement between seeds (ARI 0.993). The check costs extra clustering runs on fake data for each  $K$ .

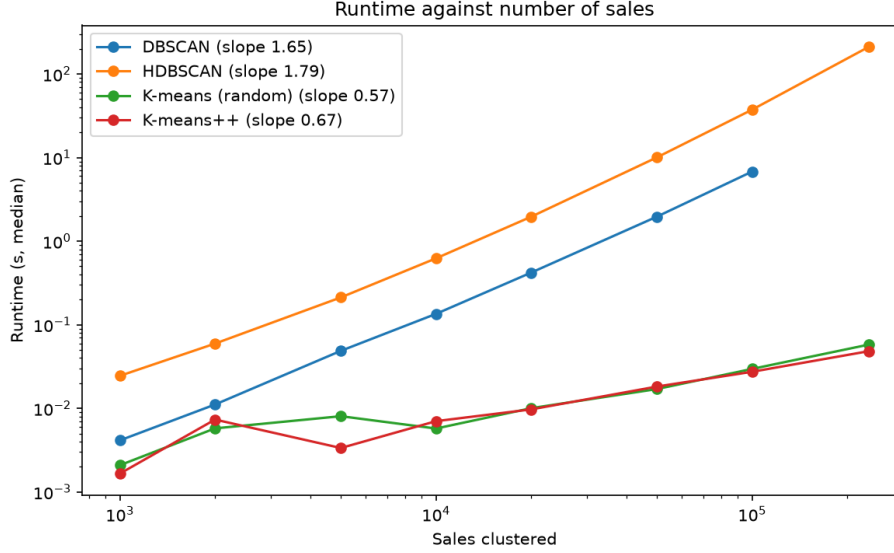
**GMM instead of K-means (D7).** [HoangAnh] K-means cuts the space halfway between two centres, which suits clusters of equal spread. On D7 the malignant class is the wider one, so the boundary falls inside it: 36 of the 212 malignant tumours join the benign cluster. With GMM that falls to 16. A GMM with one full covariance matrix per component models the spread of each class and moves the boundary accordingly: ARI rises from 0.654 to 0.774 and NMI from 0.532 to 0.661. The costs are a slower fit, 465 covariance parameters per component estimated from 569 tumours, and a lower silhouette (0.314 against 0.343), so an analyst who trusts the silhouette would reject the better result. BIC, the model’s own criterion, also selects two components with full covariance, so this fix can be chosen without labels.

**Proposed, not implemented (D6, D7).** [HoangAnh] The density methods fail on both tables because 13 and 30 standardised features leave little contrast in density. Projecting onto a few principal components before DBSCAN or HDBSCAN is the natural next step; on D7 the first two components already hold 63.2% of the variance. [All: Proposed but not implemented: e.g. ground truth for D2 from plate-boundary maps; OPTICS; approximate spectral clustering for large  $n$ .]

### 3.7 Runtime and scalability

Owner: Wei Yew, Lam

D8 is large enough to measure how each method grows. Each method ran at its chosen setting on random subsets of 1,000 to 234,294 sales (K-means from one start). Figure 8 plots the median time against size on log scales, where the slope of a line estimates the exponent  $b$  in  $t \propto n^b$ . K-means is far cheaper than either density method at every size: under 0.06 s on all 234,294 sales. Its fitted slopes (0.57 and 0.67) are below 1 only because a fixed start-up cost dominates at these sizes; each iteration is linear in  $n$ . HDBSCAN grows with slope 1.79, from 0.02 s at 1,000 sales to 212 s at 234,294. DBSCAN grows with slope 1.65, but at full size it was not run at all, because of memory rather than time. With a fixed  $\varepsilon$ , every sale has more neighbours as the data grow. Estimated from 2,000 probe points, the full data would hold about 742 million neighbour pairs, about 6 GB for the neighbour indices alone, which scikit-learn’s DBSCAN stores in full. HDBSCAN, slower at every size, did run on all sales within 16 GB. [Lam: Why D5 needed subsampling for Spectral, DBSCAN and HDBSCAN; fit times per method.] [Ky: Existing Fig. 9 runtime at  $n \leq 23,412$ , if kept.]



**Figure 8:** D8, median fit time against the number of sales, log–log, at the chosen settings; the slope of each line is in the legend. DBSCAN stops at 100,000 sales: the full data would need about 6 GB of neighbour indices.

### 3.8 Success and failure cases

Owner: each owner: one figure per case, with the cause named in the caption

#### 3.8.1 Successes

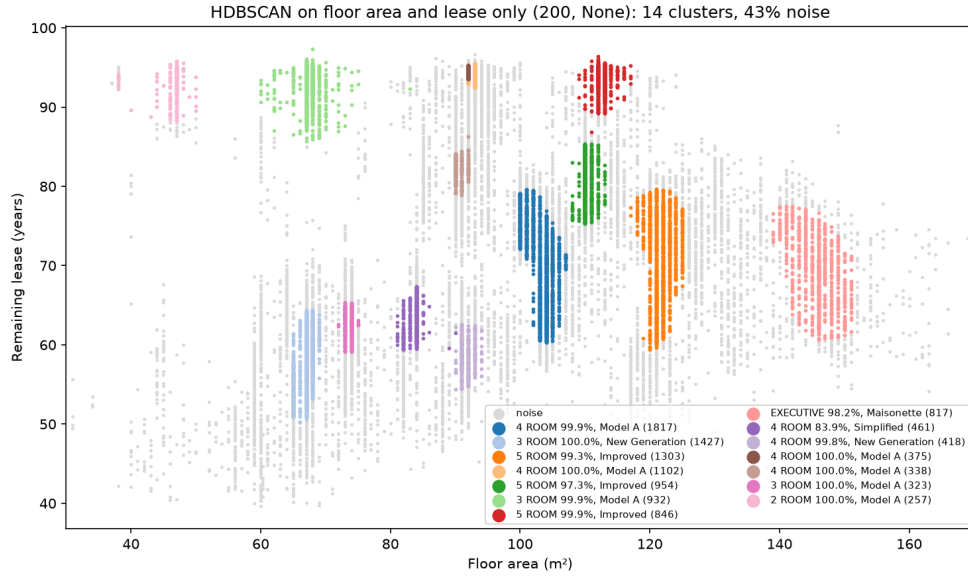
**HDBSCAN on D2: belts of varying density.** Shown in Figure 2. HDBSCAN keeps the sparse belts of the Americas (62% of earthquakes clustered, against 20% for DBSCAN) as long clusters that follow the plate boundary. It succeeds because its density threshold adapts to each cluster, the property D2 needs most.

**GMM on D7: classes of unequal spread.** [HoangAnh] Shown in Figure 10. GMM matches the diagnosis best of all methods (ARI 0.774). K-means places its boundary too far into the wide malignant class (ARI 0.654); the full covariance of each GMM component lets the boundary follow the compact benign class. Cause of success: the model allows clusters of different spread. [Lam: DBSCAN / Spectral on D1.] [Ky: K-Means++ on D4: compact classes in 7 dimensions.]

**HDBSCAN on D8, floor area and lease only: flat designs.** This feature set was chosen after seeing the main results, so we present it as an ablation finding and a sanity check, not as a tuned setting. Dropped to two features, D8 does contain dense spots: standard flat designs, built to fixed plans in particular years. HDBSCAN finds 14 clusters (43.2% noise), and 13 of them are at least 97% one flat type (Figure 9); in the median cluster one flat model accounts for 83% of sales. A simple grouping by flat model and lease would give similar groups, which is the point: the method recovers a structure we can name. ARI against flat type is still only 0.127, because each type is split into several designs and the noise counts as a cluster. A low ARI here reflects a finer partition than the labels, not a failure.

#### 3.8.2 Failures

**K-Means++ on D2: unrelated belts merged.** Shown in Figure 1. One cluster joins five belts whose centres are 2,932 to 11,449 km apart, from Iceland to the Mid-Indian Ridge. Cause:

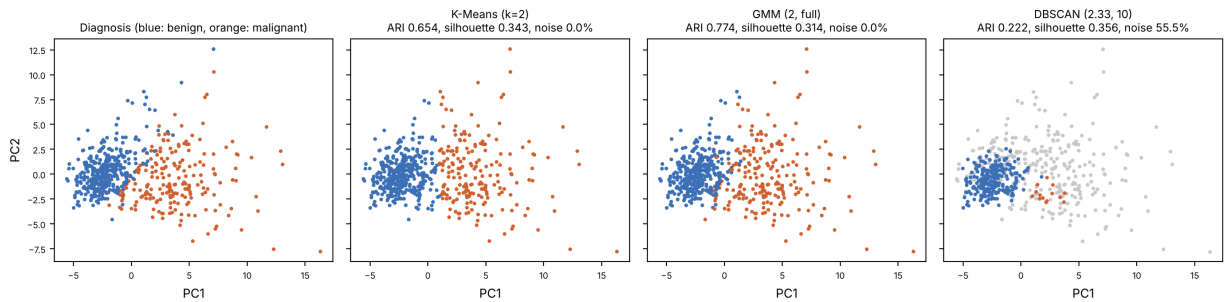


**Figure 9:** D8, success of HDBSCAN on floor area and remaining lease only (`min_cluster_size` 200; grey: noise). Each cluster is one standard flat design; the legend gives its main flat type and model. Cause of success: fixed building plans create real dense spots in these two features.

cluster shape. Compact regions cannot follow long belts.

**Density methods on D8: one mass, or storey bands.** Shown in Figures 3 and 7. At the chosen settings DBSCAN puts 97.1% of sales in one cluster. Under a cap on the largest cluster, the clusters become single storey bands. Causes: no gaps between groups, and values recorded in steps.

**DBSCAN on D7: half the data discarded.** [HoangAnh] Shown in Figure 10. The least noisy two-cluster setting keeps the dense benign core as one cluster, finds a second cluster of 14 malignant tumours and labels 55.5% of tumours noise, including 183 of the 212 malignant ones. Its silhouette (0.356) is higher than that of GMM or K-means, and its ARI is 0.222. Causes: 30 dimensions, and one class much sparser than the other.



**Figure 10:** [HoangAnh] D7 on its first two principal components (grey: noise; each cluster takes the colour of its majority class). Success of GMM against K-means: the boundary follows the compact benign class. Failure of DBSCAN: only the dense benign core is clustered. Causes: classes of unequal spread, and no density contrast in 30 dimensions.

[Lam: DBSCAN on D5.] [Ky: HDBSCAN on D4: 37% noise removes rare classes.]

### 3.9 Key factors and practical guidance

Owner: Ky

**All:** success/failure panel grid: one panel per case above

**Figure 11:** [All: caption naming the cause of each case]

[Ky: Port Sec. 7.3 (key factors) and re-rank them with evidence from all datasets: cluster shape, density variation, dimensionality, gaps between groups, scaling, initialisation, recorded steps, parameter choice.] [Ky: Port Sec. 7.4 (guidance). Add: compare every result against fake data (Wei Yew); check for values recorded in steps (Wei Yew); report noise % next to every silhouette.]

## 4 Conclusion

Owner: Ky

[Ky: Summary of findings with numbers; answer the title question.]

### Limitations

[HoangAnh] On D6 and D7, the methods that need  $k$  were given the number of classes, which an analyst would not know. The label-free criteria happen to agree here (the silhouette picks  $k = 3$  on D6 and  $k = 2$  on D7 for K-means), so the cost is small on these two tables, but both are small (178 and 569 points) and their density settings were ranked by DBCV, not by the silhouette used elsewhere.

[All: One line each: D2 has no ground truth; D8 has no location feature; oracle ARI on labelled data is optimistic; datasets up to 50 dimensions.]

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**[Ky + Hoang Anh:** Uber source: FiveThirtyEight, Uber pickups in New York City (April 2014 file); agree which copy to cite.] **[Lam:** Mall.]